Player Tracking in Sports: Multi-View Tracking and 3D Reconstruction

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1 Introduction

Vision-based tracking provides a scalable alternative to expensive sports hardware, but basketball's rapid movements, mutual occlusions, and specular floor reflections pose severe challenges. Although modern pipelines have shifted from classical background subtraction [3] to tracking-by-detection approaches [1], maintaining identity consistency during close-range interactions remains a bottleneck. This work introduces an end-to-end multi-view pipeline that addresses these issues using a domain-adapted YOLOv26 detector [5] combined with an appearance-based DeepSORT tracker [6, 8]. By leveraging multi-camera geometry and SVD-based triangulation, the system resolves 2D ambiguities and reconstructs smooth, continuous 3D trajectories across the court.

2 Detection

The detection approaches previously employed range from classical computer-vision methods to domain-adapted deep learning. As a baseline, we use a Mixture of Gaussians (MOG2) background-subtraction method [3] with CLAHE preprocessing [9]; however, it remains highly vulnerable to court-floor specularities, stationary players, and dynamic spectator backgrounds. Adopting an object-detection paradigm with a COCO-pretrained YOLOv11 model [5] substantially reduces these spatial errors by leveraging semantic features. Nonetheless, important domain gaps remain: the basketball is absent from the pre-trained taxonomy, and players and referees are both labeled as "person", preventing identity distinction.

2.1 Fine-Tuned YOLOv26 with Multi-Pass Inference

To bridge these domain gaps, we adopt the YOLOv26 architecture and fine-tune its backbone on a domain-specific annotated dataset. The model is trained to recognise each player individually, as well as the ball, so identity is resolved directly by the detector.

Because the ball occupies a very small pixel area, single-pass inference often downsamples it too aggressively for reliable localisation. We address this by employing a dual-pass inference strategy: a primary pass optimised for player detection, alongside a high-resolution pass with a permissive confidence threshold to improve ball recall. Detections from both passes are then merged into a single composite output.

To remove duplicate bounding boxes produced by conflicting identity predictions, we replace standard non-maximum suppression with a class-agnostic variant [4]. This variant suppresses detections solely on spatial overlap, ensuring a single, clean bounding box per player is forwarded to the tracker.

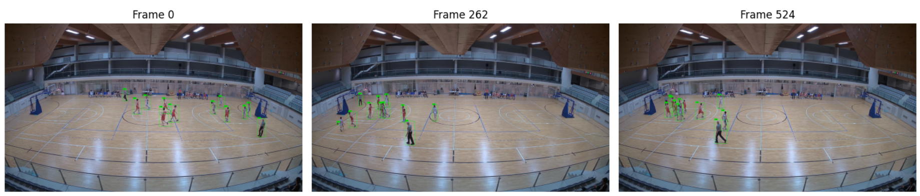

Fig. 1. Sample detection output from fine-tuned YOLOv26 on cam_13.

3 Tracking

To maintain continuous trajectories across frames, the tracking framework evolves from a purely spatial baseline to an architecture robust to appearance variation. As a baseline, we employ Simple Online and Realtime Tracking (SORT) [1], which propagates tracks using a Kalman filter and associates predicted boxes to incoming detections via the Hungarian algorithm using Intersection-over-Union (IoU) overlap. SORT's reliance on spatial overlap, however, leads to frequent identity switches when players cross paths.

3.1 DeepSORT

Our tracking framework adopts DeepSORT [6] to reduce identity swaps by augmenting the Kalman-filter motion model with appearance information. For each detection, a lightweight *OSNet* [8] produces an appearance embedding that is appended to a per-track rolling gallery. Association is performed by a matching cascade that prioritizes appearance similarity before spatial overlap.

Building on this foundation, we introduce several improvements. The **Confidence-Aware Kalman Update (NSA)** reduces state-correction jitter by scaling the measurement covariance according to detection confidence rather than treating all bounding-box updates equally. **Low-Confidence Recovery (ByteTrack)** [7] performs a second-pass association between unmatched tracks and low-confidence detections using spatial overlap, preserving continuity through partial occlusions. **Track Resurrection** addresses prolonged occlusions by retaining the appearance galleries of recently terminated tracks for a brief temporal window; when a newly detected player closely matches a retained gallery, the original identity is restored rather than initializing a new track. Finally, **Coasting** bridges short gaps by emitting a track's Kalman-predicted position while it remains unmatched, provided the prediction remains within frame bounds and does not overlap an already-assigned bounding box.

A final **Label Resolution** step unifies the detector's per-frame class predictions across each trajectory. Each track is assigned its confidence-weighted majority label; identity conflicts between tracks are resolved in favor of the higher-confidence prediction. The ball category is exempt from unique-identity arbitration.

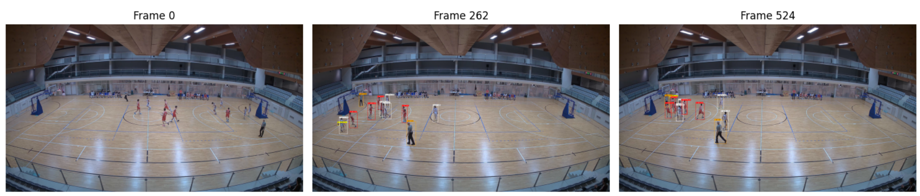

Fig. 2. Sample tracking output from DeepSORT on cam_13.

4 3D Reconstruction

Having recovered per-camera 2D trajectories, we lift them into a unified 3D scene representation shared by all three cameras. This geometric lifting uses intrinsic matrices known a priori; extrinsic parameters and lens-distortion coefficients are estimated by solving the Perspective-n-Point (PnP) problem, mapping manually annotated 2D court landmarks in the initial frame to their corresponding 3D world coordinates.

For triangulation, detections are grouped by identity and frame across cameras, then undistorted through rectification. Reconstruction proceeds separately for players and the ball due to their distinct physical constraints. Because players move on the court floor, each camera's bottom bounding-box pixel is back-projected into a ray whose intersection with the floor plane yields the ground position. This floor constraint enables single-view localisation during occlusions; multi-view estimates are produced by averaging after discarding outliers. The airborne ball is triangulated from rays from at least two cameras using the Direct Linear Transform (DLT), solved via Singular Value Decomposition (SVD). Views with high reprojection error are iteratively discarded and the point re-triangulated.

Trajectory smoothing removes residual noise from detection jitter and calibration error. Each 3D trajectory is filtered forward with a Kalman filter and refined by a Rauch–Tung–Striebel (RTS) backward pass. This bidirectional processing uses past and future observations to correct states, reject implausible spatial jumps, and split trajectories across extended gaps to avoid false extrapolation.

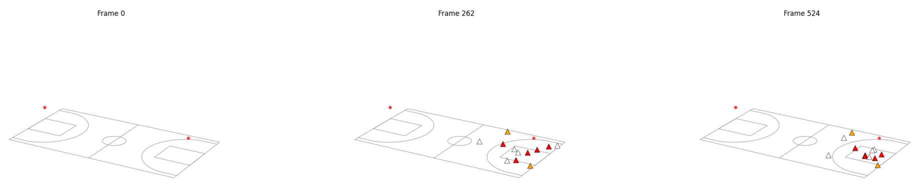

Fig. 3. Reconstructed 3D player and ball positions projected onto the court plane.

5 Evaluation Metrics

Detection is measured using precision, recall, and the F_1 score, with true positives defined at an Intersection over Union (IoU) threshold of 0.5. **Tracking** is assessed using IDF1 and HOTA [2]. IDF1 captures identity consistency, while our primary metric, HOTA, is the geometric mean of detection accuracy (DetA) and association accuracy (AssA), averaged over a range of localization thresholds. We favour HOTA over traditional metrics such as MOTA, which are biased toward detection errors and do not adequately reflect tracking continuity. **3D geometry** accuracy is evaluated using two primary metrics, reprojecting the 3D points back onto the image planes. The **reprojection error** measures geometric consistency, which reflects the quality of calibration and triangulation. The **trajectory error** quantifies tracking accuracy by comparing these 3D trajectories against ground-truth 2D data across all shared frames. We report the Average Displacement Error (ADE), the Final Displacement Error (FDE), and the Median Trajectory Error (MTE).

6 Results

This section summarises the final tracking and trajectory results. The complete detection, tracking, and geometry tables, together with per-stage qualitative outputs and all pipeline parameters, are provided in the appendix (Appendices B–D). Full video results are available in the accompanying repository.

Table 1 shows the final tracking metrics. The tracker keeps identities reasonably well, however its main limitation is detection recall rather than association. This implies that most of its errors are missed players rather than false detections, implying that a good association cannot recover it. The comparable detection (DetA) and association (AssA) scores confirm this. Across cameras,

Table 1. Tracking and trajectory results.

Camera	IDF1	HOTA	DetA	AssA
cam_13	0.569	0.434	0.469	0.405
cam_2	0.610	0.480	0.453	0.511
cam_4	0.638	0.521	0.553	0.495

Camera	ADE	FDE	MTE	Frag.
cam_13	976.8	849.8	1047.5	2
cam_2	1066.0	1279.1	1181.4	5
cam_4	1838.2	1689.1	1900.0	5

Table 2. Detection and calibration reprojection-error results.

Camera	Prec.	Rec.	F_1	IoU
cam_13	0.872	0.732	0.796	0.758
cam_2	0.835	0.761	0.796	0.757
cam_4	0.923	0.822	0.870	0.796

Camera	Mean	Median	RMSE
cam_13	51.3	28.9	87.7
cam_2	140.0	94.5	278.7
cam_4	135.0	50.0	296.2

detection recall and HOTA also rank in the same order, which confirms that detection drives the tracking quality.

The video results highlight one recurring failure mode within the tracking pipeline. First, the basketball poses a significantly greater tracking challenge than the players, serving as the primary bottleneck for the detection and tracking performance. This effect is sometimes exacerbated by background clutter (e.g., stray basketballs), which can produce spurious trajectories.

Projecting the reconstructed 3D tracks back to 2D gives the trajectory errors reported in Table 1. These errors are large, and the main cause is calibration bias: an error in the camera poses propagates through triangulation into a nearly constant positional offset. The similar ADE, FDE, and MTE values within each camera indicate a constant offset rather than growing drift, which is consistent with this bias. The fragment count (Frag.), the number of gaps in a predicted track, stays low. This shows that the trajectories remain mostly continuous, mainly thanks to the three-camera triangulation, whose redundancy recovers a position when one view misses a detection.

In the video, the reconstructed 3D trajectories look overall good enough and follow the players consistently. The large error values mainly reflect the pixel scale of the high-resolution frames, so a displacement that is large in pixels is still small relative to the image size. As in 2D, the ball is likely the main contributor to the residual 3D error.

The tracking scores and the trajectory errors do not always move together. This confirms that calibration, not tracking, is the main limiting factor, in agreement with the reprojection errors in Table 2. There, the RMSE far exceeds the median, so a few outlier frames, rather than the calibration as a whole, account for the largest errors.

References

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A Dataset and Fine-Tuning

The pipeline is evaluated on a basketball clip from the Sanbapolis facility, recorded by three synchronised cameras (cam_13, cam_2, cam_4) on a FIBA court of $28 \times 15\,\mathrm{m}$. The clip spans 525 frames at 25 fps (about 21 s). Camera intrinsics are provided, and ground-truth 2D bounding boxes are annotated on every fifth frame for evaluation.

The detector is fine-tuned on a separate Roboflow dataset from the same game, covering different actions and disjoint from the evaluation clip. Training starts from YOLOv26m weights and runs for up to 300 epochs (early stopping, patience 30) at 1280 px input and batch size 4.

B Pipeline Parameters

The following tables list the parameters used at each stage of the pipeline: MOG2 background subtraction (Table 3), YOLO detection and NMS (Table 5), tracking (Table 6), and 3D geometry (Table 4).

Table 3. MOG2 background subtraction and preprocessing.

Parameter	Value
Learning rate	−1 (adaptive)
Detect shadows	off
CLAHE clip limit	6.0
CLAHE tile grid	8×8
Min. blob area (px^2)	500
Max. blob area (px^2)	10000

Table 4. Geometry: smoothing and triangulation.

Parameter	Value
2D trajectory smoothing	
Max. interpolated gap (frames)	5
EMA weight α	0.5
Triangulation	
Ground inlier gate (mm)	600
Ball reprojection gate (px)	30
Minimum views	2
3D trajectory smoothing	
Max. join gap (frames)	5
Mahalanobis gate (χ^2)	11.345
Position noise std (mm)	30
Velocity noise std (mm/frame)	60
Measurement noise std (mm)	80

C Detailed Results

This section reports per-camera detection (Table 7), tracking (Table 8), and 3D geometry errors (Table 9).

D Qualitative Outputs

The figures below are the per-stage visual outputs produced by the pipeline notebook on cam_13.

Fig. 4. Raw input frame.

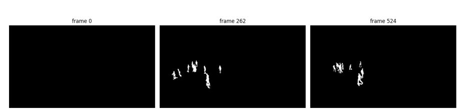

Fig. 5. MOG2 foreground mask.

Table 5. YOLO detection and NMS.

Parameter	Value
Single-pass YOLO	
Confidence threshold	0.3
Inference size (px)	640
Built-in NMS IoU	0.45
Two-pass fine-tuned	
Player inference size (px)	1280
Ball inference size (px)	1600
Ball confidence	0.2
Player low-conf band	0.15
Class-agnostic merge NMS	
IoU suppression threshold	0.75

Table 6. Tracking (SORT and DeepSORT).

Parameter	Value
SORT	
Max. IoU distance	0.8
Max. age (frames)	60
Confirmation hits	2
DeepSORT	
Max. IoU distance	0.7
Max. appearance distance	0.2
Max. age (frames)	60
Confirmation hits	2
Feature budget per track	100
Coast age (frames)	5
Coast max. IoU	0.3
Resurrection window (frames)	150
Resurrection max. distance	0.25
Appearance (OSNet)	
Input size (px)	256×128
Embedding dimension	512

Fig. 6. MOG2 detections overlaid on the frame.

Fig. 7. Pre-trained YOLOv11 detections.

Table 7. Detection results across cameras (IoU threshold = 0.5).

Method	Camera	Precision	Recall	F_1	Mean IoU
MOG2	cam_13	0.471	0.262	0.337	0.651
	cam_2	0.345	0.408	0.374	0.663
	cam_4	0.328	0.293	0.309	0.700
YOLO pre-trained	cam_13	0.834	0.602	0.699	0.748
	cam_2	0.819	0.781	0.800	0.785
	cam_4	0.927	0.807	0.862	0.808
YOLO fine-tuned	cam_13	0.872	0.732	0.796	0.758
	cam_2	0.835	0.761	0.796	0.757
	cam_4	0.923	0.822	0.870	0.796

Table 8. Tracking metrics: identity (IDP, IDR, IDF1) and HOTA (IoU threshold = 0.5).

Tracker	Camera	IDP	IDR	IDF1	HOTA	DetA	AssA	LocA
SORT	cam_13	0.660	0.514	0.590	0.438	0.484	0.398	0.797
	cam_2	0.743	0.607	0.668	0.517	0.515	0.522	0.794
	cam_4	0.768	0.605	0.679	0.542	0.575	0.514	0.829
DeepSORT	cam_13	0.588	0.533	0.569	0.434	0.469	0.405	0.769
	cam_2	0.617	0.603	0.610	0.480	0.453	0.511	0.761
	cam_4	0.670	0.605	0.638	0.521	0.553	0.495	0.798

Table 9. 3D geometry errors: reprojection of triangulated points and trajectory error of projected tracks. Mean, Median, RMSE, ADE, FDE, and MTE are in px; Acc columns are fractions. Fragment counts are 2, 5, 5 for cam_13, cam_2, cam_4.

	Reprojection error						Trajectory error		
Camera	Mean	Median	RMSE	Acc@5	Acc@10	Acc@20	ADE	FDE	MTE
cam_13	51.3	28.9	87.7	0.135	0.221	0.349	976.8	849.8	1047.5
cam_2	140.0	94.5	278.7	0.028	0.047	0.094	1066.0	1279.1	1181.4
cam_4	135.0	50.0	296.2	0.168	0.214	0.295	1838.2	1689.1	1900.0

Fig. 8. Fine-tuned YOLO merged player and ball detections.

Fig. 9. Detections after class-agnostic NMS.

Fig. 10. SORT tracks.

Fig. 11. SORT tracks after label resolution.

Fig. 12. DeepSORT tracks.

Fig. 13. DeepSORT tracks after smoothing and label resolution.

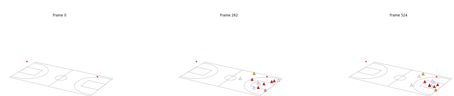

Fig. 14. Triangulated 3D reconstruction (top view).

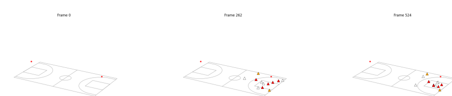

Fig. 15. Smoothed 3D reconstruction (top view).